Jiayi (Johnny) Li

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EDUCATION

Northwestern University **Evanston, IL**
B.S. Major in Computer Science & Statistics, Minor in Data Science *09/2026 - 06/2029 (Expected)*
- GPA: 4.0/4.0
- Relevant Coursework: Data Structures & Algorithms, Object Oriented Programming, Intro to Stats and Data Science

University of Toronto (Transferred) - Dean's List Scholar **Toronto, ON**
HBSc, Major in Computer Science & Quantitative Biology, Minor in Statistics *09/2025 - 08/2026*
- GPA: 3.9/4.0

RESEARCH EXPERIENCE

Radiation Dosimetry & Calorimetry-Based QA **North York, ON**
Undergraduate Researcher, Sunnybrook Hospital, under Dr. Arman Sarfehnia *01/2026 - Present*
- Built a Selenium automation pipeline in Python using WebDriverWait, XPath, and JavaScript DOM event dispatch to programmatically run SIMAC2 beam simulations on the Linax platform
- Developed a Python tool using NumPy and PyVisa that fits linear regressions to pre/post-irradiation voltage drifts to extract absorbed dose, cutting analysis from ~30 minutes to under a minute
- Built an acquisition system that samples up to 6 RTD probes on a multimeter at intervals, applying the Callendar–Van Dusen equation with per-probe calibration constants to deliver readings with live logging

Podocyte Biology & Kidney Disease Repair **Toronto, ON**
Undergraduate Researcher, University Health Network, under Dr. Moumita Barua *11/2025 - Present*
- Analyzed Col4a3 proteomics across cell line and male mouse tissue using pandas and NumPy, cross-referencing 3,414 proteins against a 23-gene fibrosis panel, and performed Western blot to validate Col4a3 antibody specificity
- Built Python visualization pipelines using matplotlib generating volcano plots with automated label placement, pathway enrichment bubble charts, GO term dot plots, and protein domain architecture maps
- Selected as a funded IMS SURP fellow through the University of Toronto's Institute of Medical Science; presented research findings via poster at SURP Research Day

CMV Infection & Environmental Heavy Metal Exposure **Evanston, IL**
Student Researcher, Northwestern University, under Dr. Hui Zhang *03/2025 - 04/2026*
- Analyzed 4 heavy metal biomarkers across ≈554 U.S. adults from NHANES 2017–2018 to quantify associations between environmental metal exposure and persistent CMV infection
- Applied and built 5+ statistical models in R including multivariable regression, Weighted Quantile Sum (WQS), GAMs, and sensitivity analyses across all metal-CMV exposure-outcome pairs to control for confounders
- Sole-authored research paper peer-reviewed and published by Asia Pacific Science Press in the Advances in Management and Intelligent Technologies (AMIT) journal

Parkinson's Disease & α-Synuclein Aggregation **Boston, MA**
Student Researcher, Harvard Medical School, under Dr. Ulf Dettmer *07/2024 - 02/2026*
- Performed 3 IncuCyte live-cell assays at Brigham & Women's Hospital, quantifying α-synuclein aggregation across 5 compounds, 3 doses, and 35+ replicates per condition
- Conducted quantitative image analysis of inclusion burden, YFP signal, and cell density across 500+ data points to identify small molecules that disaggregate lipid-rich αS inclusions
- First-author peer-reviewed publication in the Journal of High School Science (JHSS); selected and presented a poster at the 2026 American Association for Advancement of Science (AAAS) Annual Meeting

PUBLICATIONS

- Li, J. (2026). Cytomegalovirus infection in population samples: Are whole-blood levels of cadmium, mercury, selenium, and manganese associated with CMV serostatus? *Advances in Management and Intelligent Technologies*, *2*(1). https://doi.org/10.62177/amit.v2i1.1094
- Li, J & Dettmer, U. (2025). Parkinson's Disease-linked protein α-synuclein: do small-molecule inhibitors of protein aggregation also prevent its accumulation with lipids?. *Journal of High School Science*, *9*(1). https://doi.org/10.64336/001c.131827

WORK EXPERIENCE

Lendo Capital **New York City, USA**
Quantitative Analyst Intern *08/2026 - Present*

- Built a Python (pandas, yfinance) pipeline with parquet caching to ingest and clean 5 years of OHLCV data across 51 tickers, fixing a bug that let invalid rows silently inflate coverage, with per-step audit logging
- Implemented return and portfolio analytics in Python (arithmetic/log returns, rolling volatility, Sharpe ratio) across 50 equities, validated to floating-point precision against hand-calculated values
- Engineering look-ahead-safe value and momentum factor signals in Python with lagged cross-sectional ranking, building a reusable quintile-sort function for long-short backtests

Sunnybrook Hospital **North York, Canada**
Software Engineering Intern *01/2026 - 09/2026*

- Built a Selenium automation pipeline in Python using WebDriverWait, XPath, and JavaScript DOM event dispatch to programmatically run SIMAC2 beam simulations on the Linax platform
- Developed a Python tool using NumPy and PyVisa that fits linear regressions to pre/post-irradiation voltage drifts to extract absorbed dose, cutting analysis from ~30 minutes to under a minute
- Built a multithreaded PyVisa/SCPI acquisition script over GPIB applying the Callendar-Van Dusen equation across up to 6 RTD probes for real-time sub-second temperature logging

My Learning Space AI (MYLS) **Toronto, Canada**
AI Product Manager Intern *06/2025 - 05/2026*

- Conducted A/B testing and user interviews to identify key friction points, designed and iterated an optimized booking flow improving successful bookings by 9%, and drove data-backed improvements in monetization
- Created AI-generated mock prototypes and collaborated cross-functionally with engineering and design to support early-stage user research and implement data-backed product improvements
- Co-authored a 5-chapter book published by China Financial & Economic Publishing House on AI's structural impact on education and institutional decision-making in North America

PROJECTS

Morning Brief Bot – Python

- Built a fault-tolerant pipeline integrating 4 APIs (Open-Meteo, Google Calendar via OAuth2, 9 RSS/Google News feeds), with per-source exception isolation so one failed API never breaks the digest
- Wrote a regex-based keyword filter (word-boundary matching, 20+ terms) to extract relevant articles from general news feeds, with URL deduplication across overlapping queries
- Deployed as a serverless GitHub Actions cron job with encrypted repo secrets, rendering an HTML email via Jinja2 and delivering it over Gmail SMTP

Grademaster – Python

- Built a full-stack grade tracking web app in Python and Flask supporting weighted items and grouped sub-components, with a REST API that auto-saves edits on a 500ms debounce
- Engineered a what-if solver that isolates any pending assessment as a variable and solves for the exact score needed to hit a target final grade, holding all other pending items at their predicted values
- Deployed via Gunicorn on Render with atomic JSON persistence using temp-file-replace writes, requiring only pip install -r requirements.txt with no database or external storage needed

S&P 500 Market Crash Simulator – Python

- Led a 4-person team building a financial contagion simulator across 503 S&P 500 companies using a weighted graph of ~24,000 Pearson correlation edges ($r \geq 0.7$) from COVID-19 period log-returns
- Engineered a recursive contagion algorithm in Python showing crash impact across correlated neighbors, simulating user-defined crash scenarios with market cap loss estimates
- Built a full-stack interactive web dashboard using Dash and Plotly, integrating yfinance data retrieval, NetworkX graph layout computation, and NumPy correlation analysis into a single deployable pipeline

Nucleus Independent Living Services (Relating Chronic Conditions to Services) – R

- Led a 3-person team analyzing ~4,000 Nucleus client records across 6 chronic burden groups to quantify 12-month pre/post changes in hospital days, healthcare costs, and readmission rates in R
- Found clients with 5+ chronic conditions had a median 7-day hospital reduction and respiratory clients saw the largest average cost reduction at ~$22,500, identifying highest-need client groups for Nucleus services

- Performed data wrangling across 3 research questions, engineering variables for chronic burden, cost reduction, and age groupings, producing summary tables and visualizations using ggplot2

TECHNICAL SKILLS

- **Programming Languages:** Python, HTML/CSS, Javascript, R, CAD, Latex
- **Frameworks/Libraries:** Flask, Vue, React, Plotly, NumPy, Pandas, Matplotlib, SciPy, SciKit-Learn, PyTorch
- **Developer Tools:** Git, VS Code, PyCharm, Github, Render, Microsoft Offices
- **Biomedical Research:** Electrophoresis, RNAi, PCR, CRISPR, Gradient dilution, Fluorescent Microscopy